

◆ L05: Conditional Statements

The Diamond Flowchart

Neon City Cyber Academy

Module 2: Control Structures

$$\begin{aligned} 2x_1 + x_2 &= 7 \\ x_2 - 3x_3 &= \\ -2x_3 + x_4 & \\ x_3 - 3x_4 &= 1 \end{aligned}$$

Your Team

Emily Chen - *Handler*

"Until now, your code ran line by line. Today you learn to make DECISIONS - the heart of all intelligent programs."

Cole Martinez - *Tech Lead*

"Think of conditionals as diamond-shaped forks in the road. The program asks a question, then chooses a path."



Code So Far: Sequential

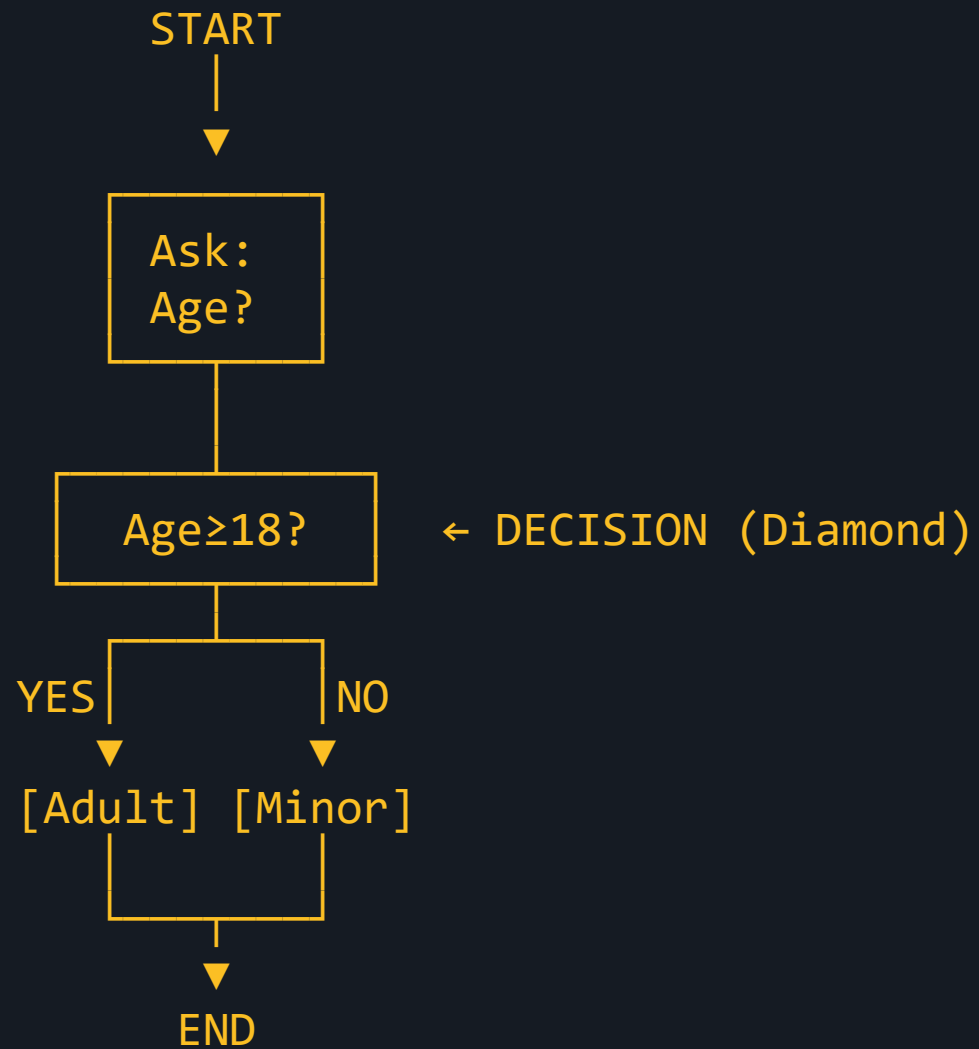
Every program we've written runs **top to bottom**:

```
print("Step 1")  
print("Step 2")  
print("Step 3")
```

Problem: No decisions! No branching! Every user gets the same experience.

Solution: Conditional statements! 💎

◆ The Diamond Flowchart



The if Statement

```
age = 20

if age >= 18:
    print("You are an adult")
    print("Access granted")
```

Syntax:

- `if` keyword
- `condition` (True or False)
- colon `:`
- `indented block` (runs if True)



Emily's Guide to if:

"The `if` statement asks a YES/NO question:"

```
if temperature > 30:  
    print("It's hot!")
```

"If the answer is YES (True), run the indented code."

"If the answer is NO (False), skip it entirely."

The condition **MUST** evaluate to Boolean: `True` or `False`

Comparison Operators

Used in conditions:

Operator	Meaning	Example	Result
<code>==</code>	Equal to	<code>5 == 5</code>	True
<code>!=</code>	Not equal	<code>5 != 3</code>	True
<code>></code>	Greater than	<code>7 > 4</code>	True
<code><</code>	Less than	<code>3 < 8</code>	True
<code>>=</code>	Greater or equal	<code>5 >= 5</code>	True
<code><=</code>	Less or equal	<code>4 <= 3</code>	False

Common Mistake: = vs ==

```
# ✗ WRONG (assignment)
if age = 18:
    print("Adult")

# ✓ CORRECT (comparison)
if age == 18:
    print("Exactly 18!")
```

= assigns, **== compares!**



The else Statement

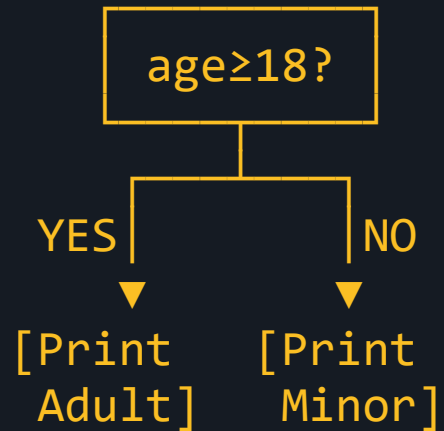
Run code when condition is **False**:

```
age = 15

if age >= 18:
    print("Adult - access granted")
else:
    print("Minor - access denied")
```

One or the other will run, never both!

💬 Cole's Flowchart:



"The diamond has TWO exits: YES path and NO path."

"The program MUST take one path - it can't take both!"



The elif Statement

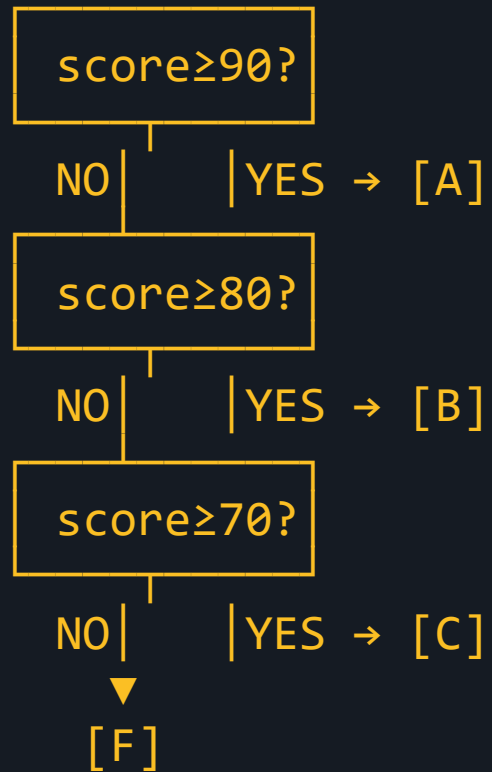
Multiple conditions:

```
score = 75

if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
else:
    print("Grade: F")
```

Checks top to bottom, stops at first True!

◆ Multi-Diamond Flowchart



Emily's elif Logic:

"`elif` means 'else if' - it's a secondary check."

```
if temperature > 30:  
    print("Hot")  
elif temperature > 20:  
    print("Warm")  
elif temperature > 10:  
    print("Cool")  
else:  
    print("Cold")
```

"Python checks each condition in ORDER. Once it finds True, it STOPS - the rest are skipped!"



Indentation Matters!

```
# ✓ CORRECT
if age >= 18:
    print("Line 1 - inside if")
    print("Line 2 - inside if")
print("Line 3 - outside if")

# ✗ WRONG (IndentationError)
if age >= 18:
print("This will crash")
```

Indentation = block membership

Use 4 spaces (or 1 tab)

Boolean Values

Conditions create Booleans:

```
is_adult = age >= 18    # True or False

if is_adult:
    print("Adult")
else:
    print("Minor")
```

You can store the result and use it later!

Logical Operators: and

Combine conditions (both must be True):

```
age = 25
has_license = True

if age >= 18 and has_license:
    print("You can drive")
else:
    print("You cannot drive")
```

Truth table:

- True and True → True
- True and False → False
- False and True → False
- False and False → False

Logical Operators: or

At least ONE must be True:

```
day = "Saturday"

if day == "Saturday" or day == "Sunday":
    print("It's the weekend!")
else:
    print("It's a weekday")
```

Truth table:

- True or True → True
- True or False → True
- False or True → True
- False or False → False

Logical Operators: not

Reverse the condition:

```
logged_in = False

if not logged_in:
    print("Please log in")
else:
    print("Welcome back!")
```

Truth table:

- `not True` → False
- `not False` → True



Cole's Compound Conditions:

```
score = 85
attendance = 95

if score >= 80 and attendance >= 90:
    print("Excellent student!")
elif score >= 70 or attendance >= 90:
    print("Good student!")
else:
    print("Needs improvement")
```

"You can combine AND, OR, NOT in complex ways. Use parentheses for clarity!"

🎮 Example: Access Control

```
username = input("Username: ")
password = input("Password: ")

if username == "admin" and password == "1234":
    print("✅ Access granted")
    print("Welcome, admin!")
else:
    print("❌ Access denied")
    print("Invalid credentials")
```

12 34 Example: Number Classifier

```
num = int(input("Number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Three-way decision with elif !



Example: Temperature Advisory

```
temp = int(input("Temperature (°C): "))

if temp >= 35:
    print("Extreme heat warning!")
elif temp >= 25:
    print("Pleasant weather")
elif temp >= 10:
    print("Cool - wear a jacket")
else:
    print("Cold - bundle up!")
```



Challenge: Grade Calculator

```
score = int(input("Score (0-100): "))

if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
elif score >= 70:
    grade = "C"
elif score >= 60:
    grade = "D"
else:
    grade = "F"

print(f"Your grade: {grade}")
```



Challenge: Password Strength

```
password = input("Enter password: ")
length = len(password)

if length >= 12 and any(c.isdigit() for c in password):
    print("Strong password ✅")
elif length >= 8:
    print("Moderate password ⚠️")
else:
    print("Weak password ❌")
```




Common Errors

Error #1: Missing colon

```
if age >= 18      # ✗ SyntaxError
    print("Adult")
```

Error #2: Wrong indentation

```
if age >= 18:
print("Adult")    # ✗ IndentationError
```

Error #3: Using = instead of ==

```
if age = 18:      # ✗ SyntaxError
    print("18")
```

✓ Lesson Complete!

You Mastered:

- ✓ if/elif/else statements
- ✓ Diamond flowchart visualization
- ✓ Comparison operators (==, !=, >, <, >=, <=)
- ✓ Logical operators (and, or, not)
- ✓ Boolean values (True/False)
- ✓ Proper indentation
- ✓ Multi-way decisions

Your Assignment

Create a **Vending Machine Program**:

1. Show menu with prices
2. Get user's selection (1-4)
3. Get amount paid
4. Use if/elif/else to:
 - Check if selection is valid
 - Check if payment is enough
 - Calculate change
 - Dispense item

Combine everything you've learned!



Next Mission!

L06: Nested Conditionals

Security Clearance Levels - if statements INSIDE if statements!